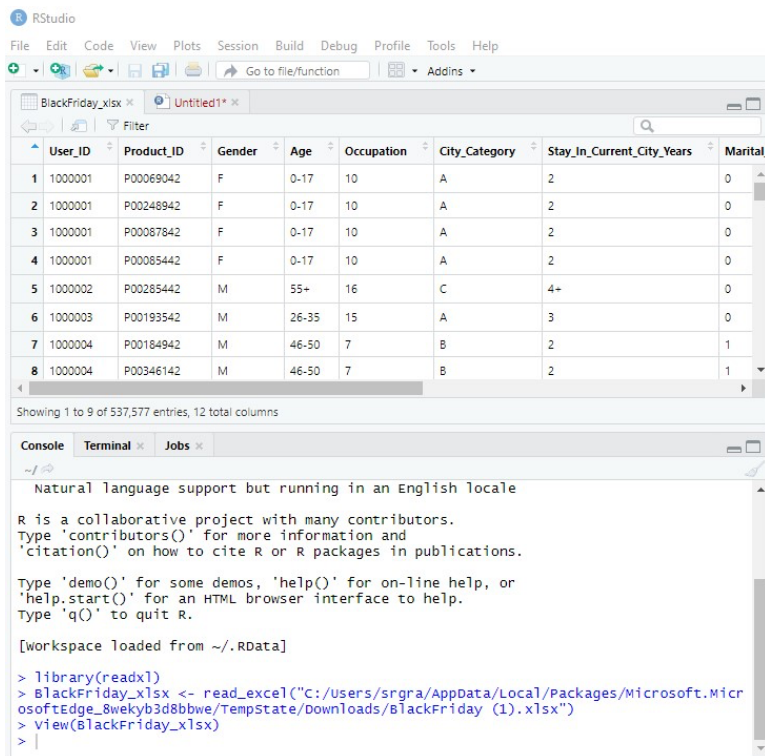


Week 3 Sean Graham

1.

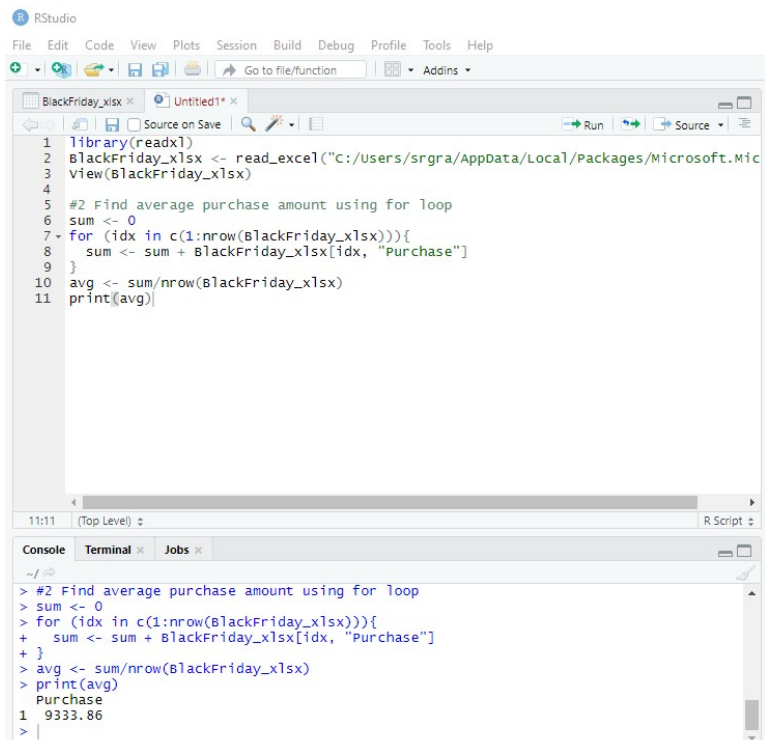


RStudio interface showing a data table with columns: User_ID, Product_ID, Gender, Age, Occupation, City_Category, Stay_In_Current_City_Years, and Marital_Status. The table displays 8 rows of data. Below the table, the console shows the R script used to load the data and view it.

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_Years	Marital_Status
1	1000001	P00069042	F	0-17	10	A	2	0
2	1000001	P00248942	F	0-17	10	A	2	0
3	1000001	P00087842	F	0-17	10	A	2	0
4	1000001	P00085442	F	0-17	10	A	2	0
5	1000002	P00285442	M	55+	16	C	4+	0
6	1000003	P00193542	M	26-35	15	A	3	0
7	1000004	P00184942	M	46-50	7	B	2	1
8	1000004	P00346142	M	46-50	7	B	2	1

```
> library(readxl)
> BlackFriday_xlsx <- read_excel("C:/Users/srgra/AppData/Local/Packages/Microsoft.Micro
osoftEdge_8wekyb3d8bbwe/Tempstate/Downloads/BlackFriday (1).xlsx")
> view(BlackFriday_xlsx)
> |
```

2.

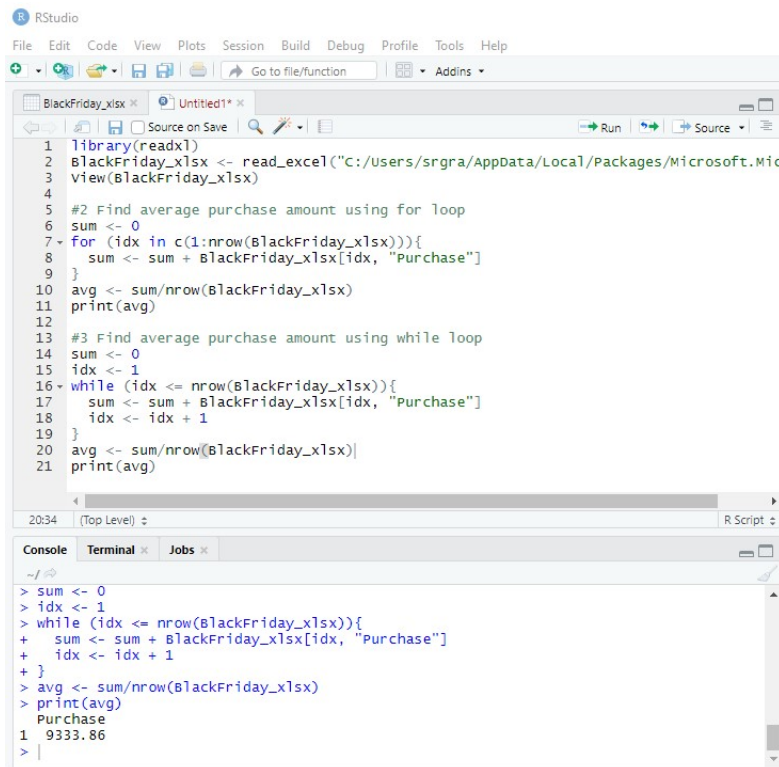


RStudio interface showing R code in the editor and the console output. The code calculates the average purchase amount using a for loop.

```
1 library(readxl)
2 BlackFriday_xlsx <- read_excel("C:/Users/srgra/AppData/Local/Packages/Microsoft.Micro
3 view(BlackFriday_xlsx)
4
5 #2 Find average purchase amount using for loop
6 sum <- 0
7 for (idx in c(1:nrow(BlackFriday_xlsx))) {
8   sum <- sum + BlackFriday_xlsx[idx, "Purchase"]
9 }
10 avg <- sum/nrow(BlackFriday_xlsx)
11 print(avg)
```

```
> #2 Find average purchase amount using for loop
> sum <- 0
> for (idx in c(1:nrow(BlackFriday_xlsx))) {
+   sum <- sum + BlackFriday_xlsx[idx, "Purchase"]
+ }
> avg <- sum/nrow(BlackFriday_xlsx)
> print(avg)
      Purchase
1  9333.86
> |
```

3.



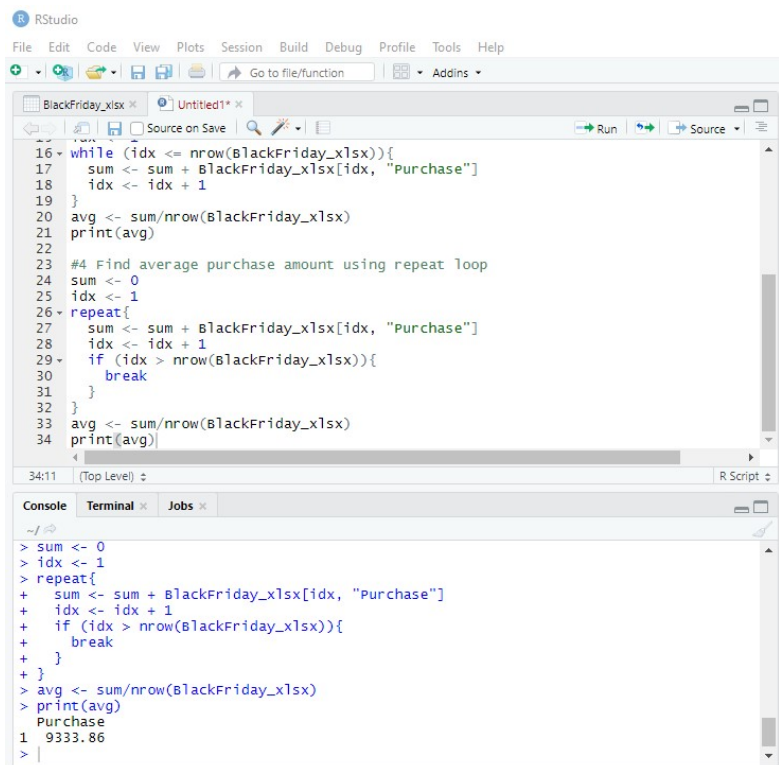
```
library(readxl)
BlackFriday_xlsx <- read_excel("C:/Users/srga/AppData/Local/Packages/Microsoft.Mic
view(BlackFriday_xlsx)

#2 Find average purchase amount using for loop
sum <- 0
for (idx in c(1:nrow(BlackFriday_xlsx))){
  sum <- sum + BlackFriday_xlsx[idx, "Purchase"]
}
avg <- sum/nrow(BlackFriday_xlsx)
print(avg)

#3 Find average purchase amount using while loop
sum <- 0
idx <- 1
while (idx <= nrow(BlackFriday_xlsx)){
  sum <- sum + BlackFriday_xlsx[idx, "Purchase"]
  idx <- idx + 1
}
avg <- sum/nrow(BlackFriday_xlsx)
print(avg)
```

```
> sum <- 0
> idx <- 1
> while (idx <= nrow(BlackFriday_xlsx)){
+   sum <- sum + BlackFriday_xlsx[idx, "Purchase"]
+   idx <- idx + 1
+ }
> avg <- sum/nrow(BlackFriday_xlsx)
> print(avg)
Purchase
1 9333.86
> |
```

4.

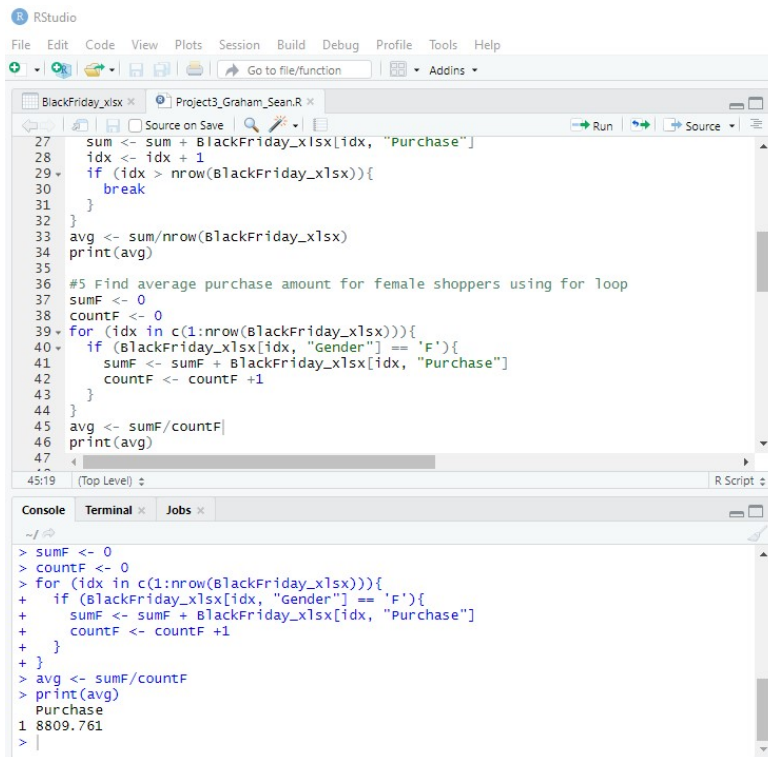


```
16- while (idx <= nrow(BlackFriday_xlsx)){
17   sum <- sum + BlackFriday_xlsx[idx, "Purchase"]
18   idx <- idx + 1
19 }
20 avg <- sum/nrow(BlackFriday_xlsx)
21 print(avg)

#4 Find average purchase amount using repeat loop
sum <- 0
idx <- 1
26- repeat{
27   sum <- sum + BlackFriday_xlsx[idx, "Purchase"]
28   idx <- idx + 1
29-   if (idx > nrow(BlackFriday_xlsx)){
30     break
31   }
32 }
33 avg <- sum/nrow(BlackFriday_xlsx)
34 print(avg)
```

```
> sum <- 0
> idx <- 1
> repeat{
+   sum <- sum + BlackFriday_xlsx[idx, "Purchase"]
+   idx <- idx + 1
+   if (idx > nrow(BlackFriday_xlsx)){
+     break
+   }
+ }
> avg <- sum/nrow(BlackFriday_xlsx)
> print(avg)
Purchase
1 9333.86
> |
```

5.



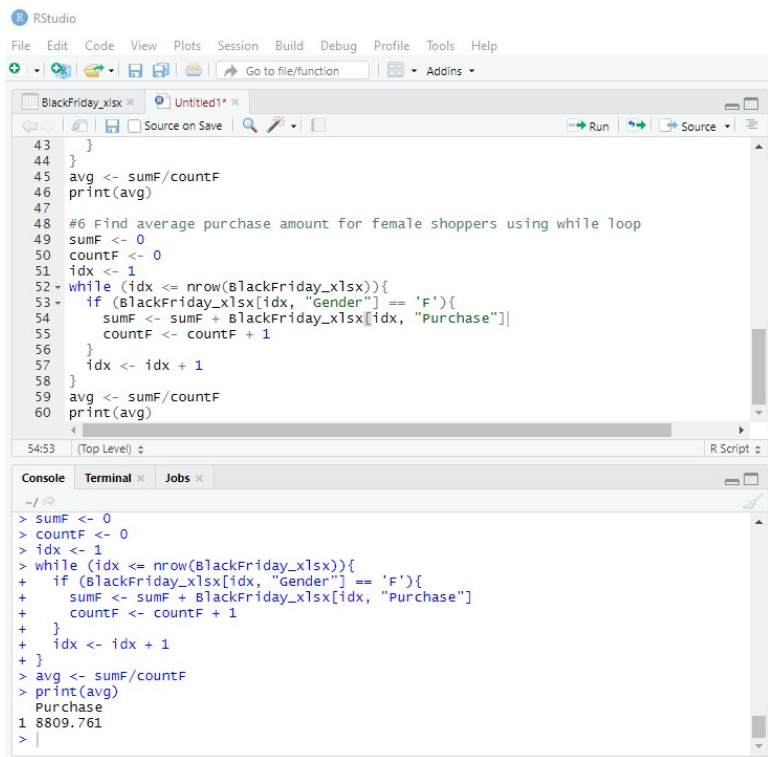
The screenshot shows the RStudio interface with a script editor and a console. The script editor contains R code that calculates the average purchase amount for female shoppers using a for loop. The console shows the execution of the script, including the initialization of variables and the final output of the average purchase amount.

```
27 sum <- sum + BlackFriday_xlsx[idx, "Purchase"]
28 idx <- idx + 1
29 if (idx > nrow(BlackFriday_xlsx)){
30   break
31 }
32 }
33 avg <- sum/nrow(BlackFriday_xlsx)
34 print(avg)
35
36 #5 Find average purchase amount for female shoppers using for loop
37 sumF <- 0
38 countF <- 0
39 for (idx in c(1:nrow(BlackFriday_xlsx))){
40   if (BlackFriday_xlsx[idx, "Gender"] == 'F'){
41     sumF <- sumF + BlackFriday_xlsx[idx, "Purchase"]
42     countF <- countF + 1
43   }
44 }
45 avg <- sumF/countF
46 print(avg)
47
```

Console output:

```
> sumF <- 0
> countF <- 0
> for (idx in c(1:nrow(BlackFriday_xlsx))){
+   if (BlackFriday_xlsx[idx, "Gender"] == 'F'){
+     sumF <- sumF + BlackFriday_xlsx[idx, "Purchase"]
+     countF <- countF + 1
+   }
+ }
> avg <- sumF/countF
> print(avg)
Purchase
1 8809.761
>
```

6.



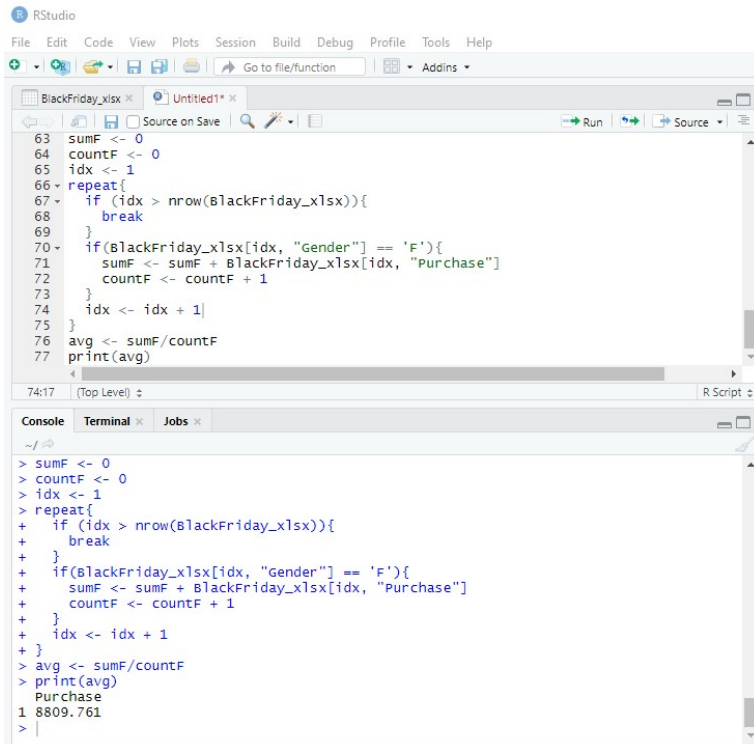
The screenshot shows the RStudio interface with a script editor and a console. The script editor contains R code that calculates the average purchase amount for female shoppers using a while loop. The console shows the execution of the script, including the initialization of variables and the final output of the average purchase amount.

```
43 }
44 }
45 avg <- sumF/countF
46 print(avg)
47
48 #6 Find average purchase amount for female shoppers using while loop
49 sumF <- 0
50 countF <- 0
51 idx <- 1
52 while (idx <= nrow(BlackFriday_xlsx)){
53   if (BlackFriday_xlsx[idx, "Gender"] == 'F'){
54     sumF <- sumF + BlackFriday_xlsx[idx, "Purchase"]
55     countF <- countF + 1
56   }
57   idx <- idx + 1
58 }
59 avg <- sumF/countF
60 print(avg)
```

Console output:

```
> sumF <- 0
> countF <- 0
> idx <- 1
> while (idx <= nrow(BlackFriday_xlsx)){
+   if (BlackFriday_xlsx[idx, "Gender"] == 'F'){
+     sumF <- sumF + BlackFriday_xlsx[idx, "Purchase"]
+     countF <- countF + 1
+   }
+   idx <- idx + 1
+ }
> avg <- sumF/countF
> print(avg)
Purchase
1 8809.761
>
```

7.



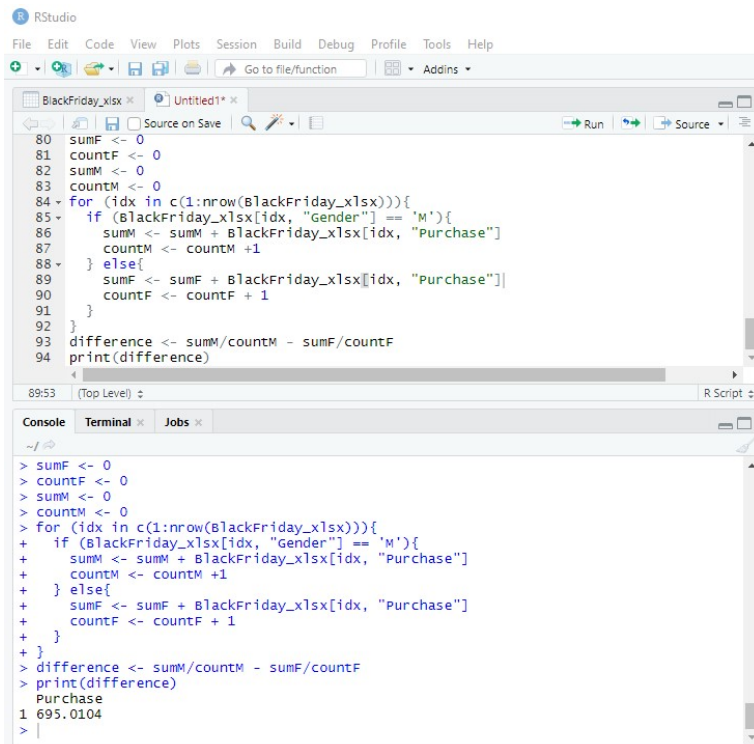
RStudio interface showing R code for calculating the average purchase amount for females using a repeat loop. The code is in a script editor with line numbers 63 to 77. The console shows the execution output.

```
63 sumF <- 0
64 countF <- 0
65 idx <- 1
66 repeat{
67   if (idx > nrow(BlackFriday_xlsx)){
68     break
69   }
70   if(BlackFriday_xlsx[idx, "Gender"] == 'F'){
71     sumF <- sumF + BlackFriday_xlsx[idx, "Purchase"]
72     countF <- countF + 1
73   }
74   idx <- idx + 1
75 }
76 avg <- sumF/countF
77 print(avg)
```

Console output:

```
> sumF <- 0
> countF <- 0
> idx <- 1
> repeat{
+   if (idx > nrow(BlackFriday_xlsx)){
+     break
+   }
+   if(BlackFriday_xlsx[idx, "Gender"] == 'F'){
+     sumF <- sumF + BlackFriday_xlsx[idx, "Purchase"]
+     countF <- countF + 1
+   }
+   idx <- idx + 1
+ }
> avg <- sumF/countF
> print(avg)
      Purchase
1 8809.761
>
```

8.



RStudio interface showing R code for calculating the difference in average purchase amounts between males and females using a for loop. The code is in a script editor with line numbers 80 to 94. The console shows the execution output.

```
80 sumF <- 0
81 countF <- 0
82 sumM <- 0
83 countM <- 0
84 for (idx in c(1:nrow(BlackFriday_xlsx))){
85   if (BlackFriday_xlsx[idx, "Gender"] == 'M'){
86     sumM <- sumM + BlackFriday_xlsx[idx, "Purchase"]
87     countM <- countM + 1
88   } else{
89     sumF <- sumF + BlackFriday_xlsx[idx, "Purchase"]
90     countF <- countF + 1
91   }
92 }
93 difference <- sumM/countM - sumF/countF
94 print(difference)
```

Console output:

```
> sumF <- 0
> countF <- 0
> sumM <- 0
> countM <- 0
> for (idx in c(1:nrow(BlackFriday_xlsx))){
+   if (BlackFriday_xlsx[idx, "Gender"] == 'M'){
+     sumM <- sumM + BlackFriday_xlsx[idx, "Purchase"]
+     countM <- countM + 1
+   } else{
+     sumF <- sumF + BlackFriday_xlsx[idx, "Purchase"]
+     countF <- countF + 1
+   }
+ }
> difference <- sumM/countM - sumF/countF
> print(difference)
      Purchase
1 695.0104
>
```